Zhengqian Jin

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Fields of Interest: Frontier Battery Technology (AFLMB, SEI Evolution); Advanced Material Design (2D High-Entropy Materials); First-Principles Calculations (DFT); AI/ML-Driven Discovery; Scientific Visualization & Tooling.

Education

2023 – Present **Ph.D. in Electronic Science and Technology**
Xi'an Jiaotong University (XJTU), Xi'an, China
Advisors: Prof. Kai Xi & Prof. Shujiang Ding

2019 – 2023 **B.S. in Applied Chemistry**
Tongji University, Shanghai, China
Honors: Outstanding Bachelor's Thesis of Tongji University; Second Prize for Outstanding University Student Paper in Shanghai.

Publications

‡ *indicates equal contribution,* @ *indicates corresponding author*

First & Co-first Author

1. **Z. Jin**[@], K. Xi. "Adapting to Information Overload: A Proposed Framework for Digital Research." *Matter*, 9 (2026) 102777.
2. **Z. Jin**, Z. Cao, L. Jin, S. Ding, K. Xi. "High-Entropy Layered Hydroxides: Pioneering Synthesis, Mechanistic Insights, and Multifunctional Applications in Sustainable Energy and Biomedicine." *Nano-Micro Letters*, 18 (2026) 200.
3. Z. Cao[‡], **Z. Jin**[‡], W. Li[‡], P. Li, Y. He, K. Jia, C. Liu, N. Li, M. Xu, W. Tang, W. Xue, R.V. Kumar, S. Ding, K. Xi[@]. "Intramolecular Polarization–Mediated Solvation and Interphase Engineering for Low–Temperature High–Voltage Lithium Metal Batteries." *Advanced Materials* (2026) 74260.
4. L. Jin[‡], **Z. Jin**[‡], T. Deng[‡], P. Liu, J. Tian, R. Gao, ... & K. Xi. "Boosting Solid–Solid Conversion Kinetics via Electron-Pinned Interface Engineering for High-Energy-Density Li–S Batteries." *Advanced Materials*, (2026) e73517.
5. T. Deng[‡], **Z. Jin**[‡], L. Jin, H. Luo, J. Wang, J. Gao, ... & K. Xi. "Relay catalyst for accelerating lithium polysulfide conversion kinetics and long-life lithium sulfur batteries." *Nano Energy*, 138 (2025) 110896.
6. M. Xu[‡], X. Wang[‡], **Z. Jin**[‡], et al. "Decoupling the roles of grain boundary strength and grain size hidden in grain-level electro-chemo-mechanical failure of solid-state electrolyte." *Journal of*

Energy Chemistry, 101 (2025) 685-691.

Co-authored Publications

7. X. Chen, X. Cui, J. Wang, Y. Chen, C. Lin, Z. Wang, J. Liu, S. Liu, **Z. Jin**, J. Wang, K. Xi, W. Yan. "A Hierarchically Porous Multifunctional Current Collector for High-Performance Li–S Batteries." *Advanced Functional Materials*, (2026) e75650.

8. H. Li, H. Jin, T. Deng, C. Yang, **Z. Jin**, Y. Zhang, ... & K. Xi. "Anion-Defect Engineering in Transition Metal Compounds for Lithium-Sulfur Batteries: Current Progress, Mechanistic Insights, and Future Directions." *Advanced Energy Materials*, (2025) e04213.

9. C. Liu, **Z. Jin**, X. Xu, Z. Cao, Y. Wang, Z. Sun, ... & K. Xi. "Accelerating Near-Electrode Li^+ Transport via High-Electronegativity Substrates for Sustainable Fast-Charging Lithium Metal Batteries." *Advanced Functional Materials*, (2025) e24392.

10. C. Liu, W. Li, Z. Wang, **Z. Jin**, T. Deng, Z. Yang, ... & K. Xi. "Crystal Facet-Engineered Anion Regulation Enables Fast-Charging Stability in Lithium Metal Batteries." *Angewandte Chemie International Edition*, (2025) e202512742.

11. H. Jin, H. Li, T. Deng, C. Yang, K. Pan, **Z. Jin**, ... & K. Xi. "Modulating carrier-catalyst heterointerfaces to boost catalytic polysulfide conversion in lithium-sulfur batteries." *Energy Materials*, 5 (2025) 500107.

12. T. Deng, J. Wang, H. Zhao, **Z. Jin**, L. Jin, X. Men, ... & K. Xi. "Dynamically Regulating Polysulfide Degradation via Organic Sulfur Electrolyte Additives in Lithium-Sulfur Batteries." *Advanced Energy Materials*, 14 (2024) 2402319.

13. J. Hu, L. Huang, W. Liu, **Z. Jin**, X. Wang, X. Xing. "Adsorption of O_2 on cationic gold clusters: Analyses on bonding strength and extent of activation." *Chinese Journal of Chemical Physics*, 37 (2024) 351–360.

14. J. Hu, J. Ma, **Z. Jin**, W. Liu, L. Huang, X. Wang@ and X. Xing@. "Reactivity of cationic silver clusters with O_2: a probe of interplay between clusters' geometric and electronic structures." *Physical Chemistry Chemical Physics*, 26 (2024) 7407-7415.

Patents

1. K. Xi, **Z. Jin**, et al. "A modified separator with catalyst for lithium-sulfur batteries, preparation method, and application." CN202511309541.7.

2. K. Xi, T. Deng, **Z. Jin**, et al. "A relay catalyst for lithium-sulfur batteries, preparation method, and application." CN202510321209.6.

3. K. Xi, C. Liu, Y. Deng, **Z. Jin**, et al. "An MXene material, preparation method, and application." CN202510539047.3.

4. H. Li, K. Xi, K. Pan, H. Jin, C. Yang, S. Yin, X. Wang, T. Deng, **Z. Jin**, et al. "A composite cathode material for Li-S batteries based on 1T phase MoS_2/graphene composites." CN202510380459.7.

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1. K. Xi, **Z. Jin**. *AtomFeed*: Battery Literature Intelligent Push Software [V1.0]. Chinese Software Copyright Reg. No. 2026SR0729645. Copyright holder: Xi'an Jiaotong University. Registered July 2026.

Selected Projects & Service

2024 – Present	**SHOCK Kit (Hardware Development)** Developing an intelligent mixed vibration control suite for standardized safety testing of pouch cells.
2022 – Present	**PaperBot (@EKL_Batteries)** Created and maintain an automated Python-driven bot on X (Twitter) to parse and share the latest battery field research.
2019 – Present	**Public Relations & Outreach** Served as a core member of the Propaganda Department at Tongji University and Publicity Committeeman for Ph.D. Class B2391 at XJTU.

Honors & Awards

2024	Outstanding Student Backbone & Outstanding Volunteer Service Team, XJTU
2023	Third Prize, Shaanxi Province University Student Microstructure Photography Competition
2022	Bronze Award, 8th China International "Internet+" Innovation and Entrepreneurship Competition
2019 – 2023	National Encouragement Scholarship (Multiple times)

Skills & Interests

Computational	First-Principles Calculations (DFT), Machine Learning for Materials Science
Programming	Python (PyTorch, Scikit-learn, Pandas, NumPy), MATLAB, Origin
Languages	Chinese (Native), English (Fluent), Spanish (Conversational)
Interests	Science visualization, long-distance running (half-marathon), football, UFC, NFL